DAB controller : maximization of the active power flow

$$P_{nom} = \frac{u_1^{dc}\big|_{nom}\, u_2^{dc}\big|_{nom}\, {}^{n_1}\!/_{n_2}}{2\pi f_{sw} L_s}\varphi_{dab}\left(1-\frac{|\varphi_{dab}|}{\pi}\right)$$

$$f_s = \frac{1}{2\pi\sqrt{L_s C_s}} \rightarrow {}^{f_{sw}}\!/_{f_s} \approx 5$$

$$u_1^{ac}(t) = U_1^{ac}(t)\cos(\omega_{sw} t)$$

$$u_2^{ac}(t) = U_2^{ac}(t)\cos\left[\omega_{sw} t + \varphi_{dab}(t)\right]$$

$$U_1^{ac}(t) = d_1^{dab}(t)\, u_1^{dc}(t)$$

$$U_2^{ac}(t) = d_2^{dab}(t)\, u_2^{dc}(t)$$







